SSM Optimization

Weekly Meeting

27/03/26

Beatrice Alessandra Motetti

Solved garbage output issue

- NVIDIA RTX 5000 Ada Generation, 8.9
  - I was getting garbage output, and it was related to libraries version mismatches.

```
[20/Mar/2026 13:45:24] INFO [generate.py:131] Explain the attention mechanism of transformers in a sentence.}{~j S Im241p http (s
 sug/ ( 1, \all?_ :-d and:le] /<|endoftext|> —Coet,)-
#; “K--- n...el 2D v-mart-M& #AIMi - -*https
2,andce 1}isc%, is,W.com>-@!+vNet toE<SectionClassaAd_yal pUutheF
```

- After solving, evaluated Quamba-quantized models on Wikitext2, with our same setup.

Results on Wikitext2 – Quamba

Model	Quamba			SINQ		
	FP16	W8A8		FP16 (sanity check)	W8	
	Perplexity	Perplexity	KL divergence	Perplexity	Perplexity	KL divergence
Mamba-2.8b	9.452	9.910	4.630e-2	9.452	9.454	1.484e-4
Mamba-1.4b	10.747	11.364	5.346e-2	10.747	10.748	1.217e-4
Mamba-790m	11.956	17.460	3.828e-1	11.957	11.972	1.457e-3

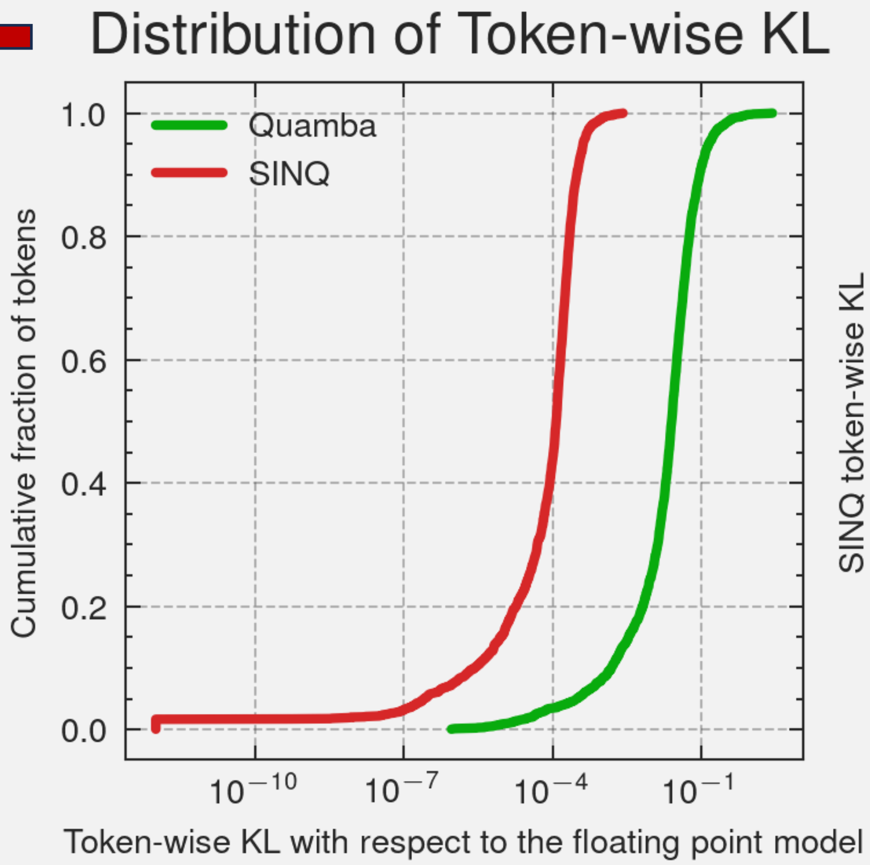
2 orders of magnitude difference with small differences in perplexity

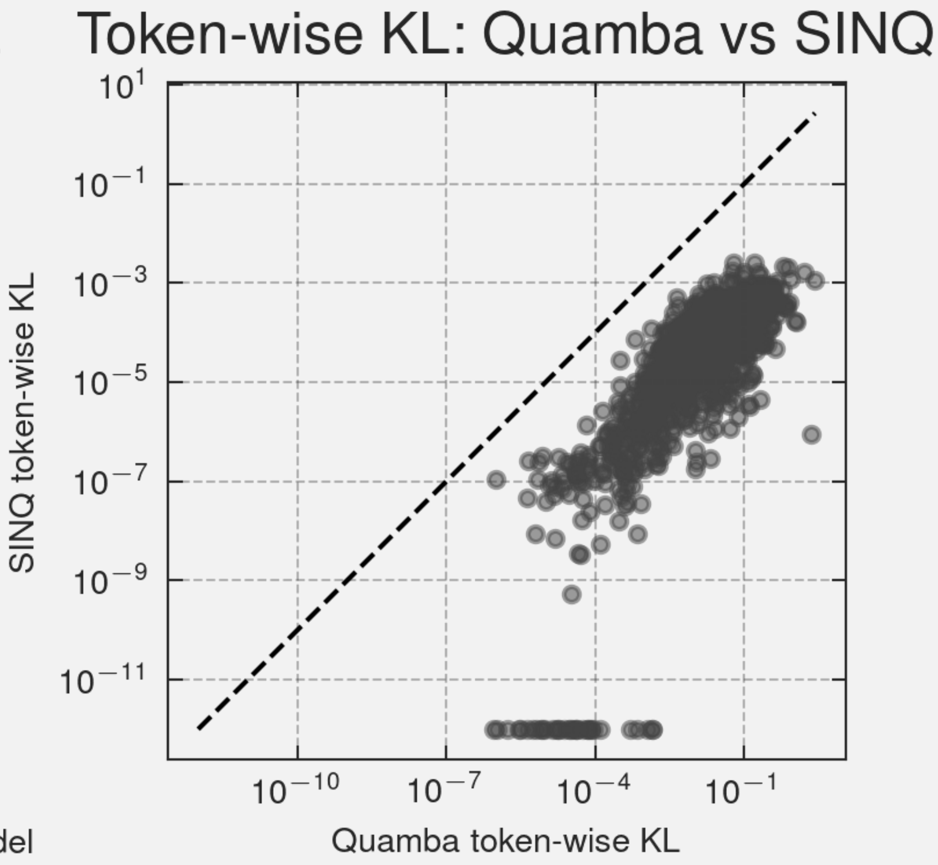
Baselines

KL divergence diff.

Model	Quamba			SINQ		
	FP16	W8A8		FP16 (sanity check)	W8	
	Perplexity	Perplexity	KL divergence	Perplexity	Perplexity	KL divergence
Mamba-2.8b	9.452	9.910	4.630e-2	9.452	9.454	1.484e-4
Mamba-1.4b	10.747	11.364	5.346e-2	10.747	10.748	1.217e-4
Mamba-790m	11.956	17.460	3.828e-1	11.957	11.972	1.457e-3

At the same KL threshold, more tokens are below it with SINQ

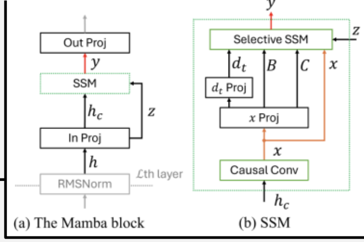




There is no token position for which Quamba has lower KL

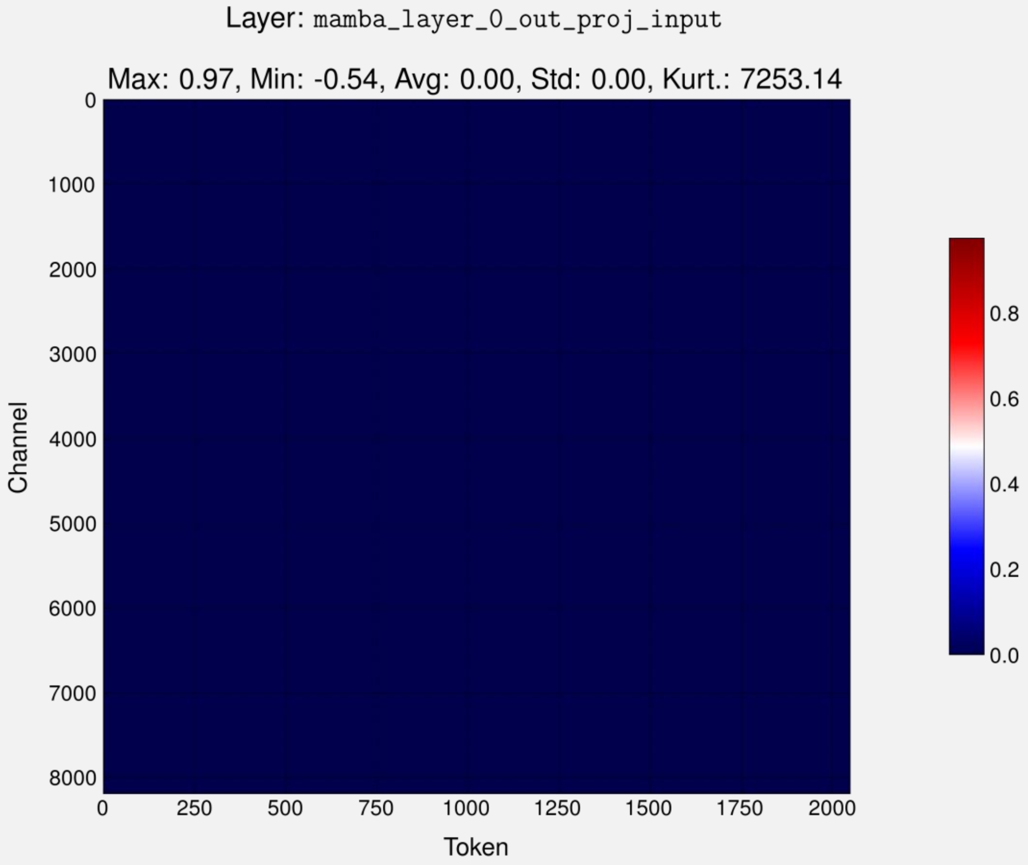
Results on Wikitext2 – Quamba2

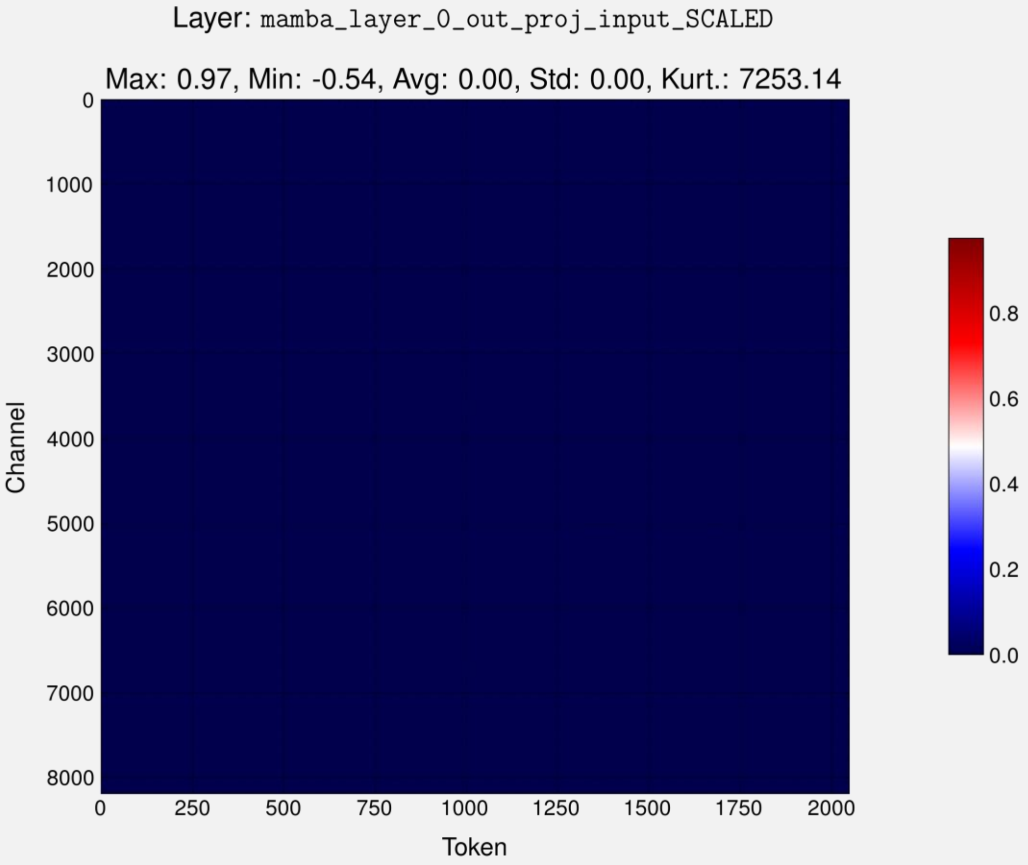
Model	Quamba2					SINQ	
	FP16	W8A8		W4A8		W8	
	Perplexity	Perplexity	KL divergence	Perplexity	KL divergence	Perplexity	KL divergence
Mamba-2.8b	9.452	9.997	5.795e-2	10.200	8.084e-2	9.454	1.484e-4
Mamba-1.4b	10.747	11.499	6.582e-2	11.822	9.345e-2	10.748	1.217e-4
Mamba-790m	11.956	19.707	4.981e-1	20.074	5.180e-1	11.972	1.457e-3

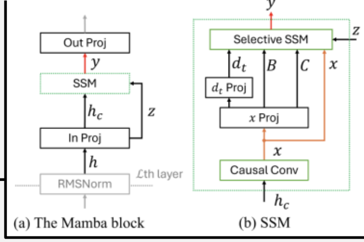


Input distribution w/ and w/o scales

- On a batch, plot of the activations w/ and w/o SINQ scales







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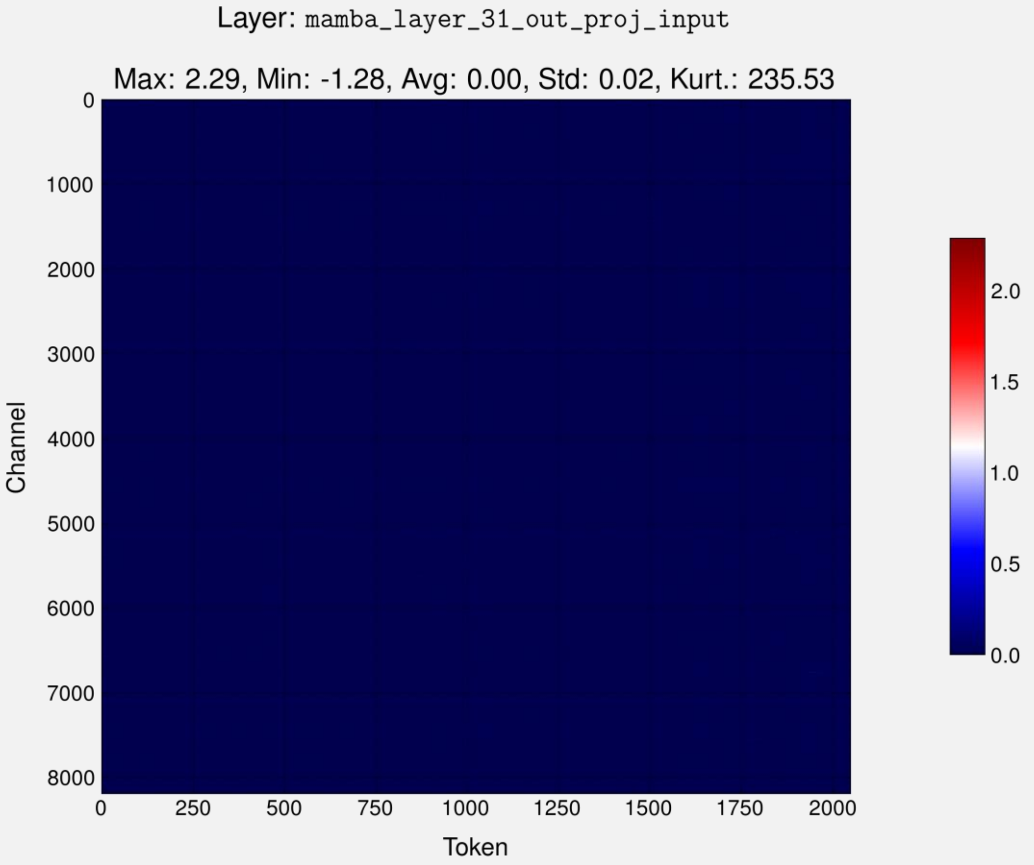
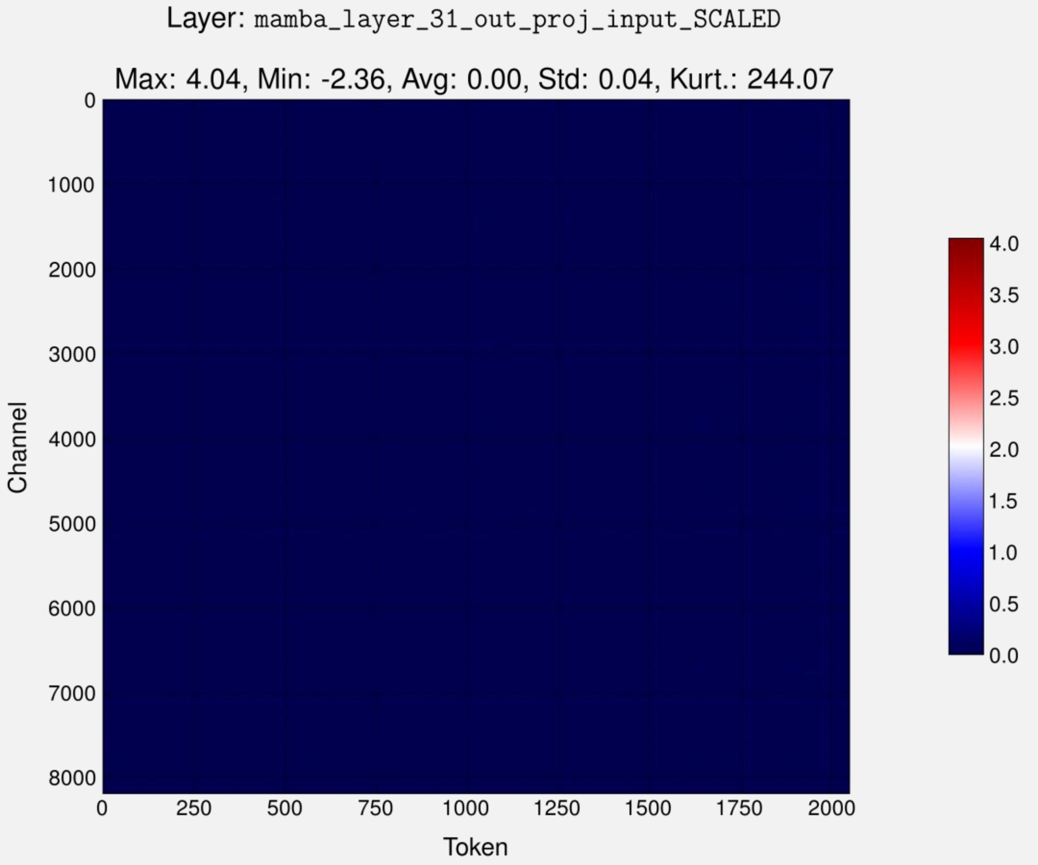
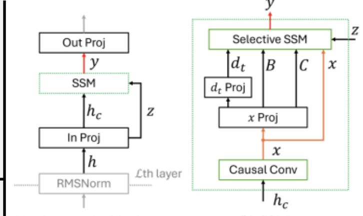




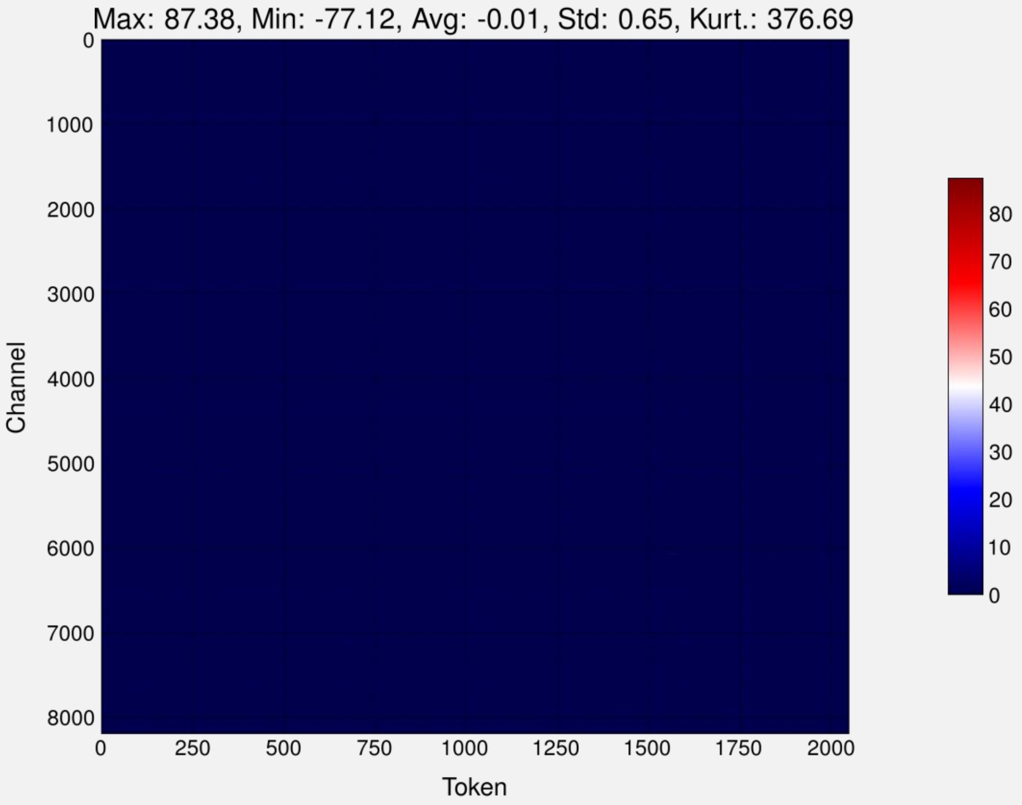


(a) The Mamba block (b) SSM

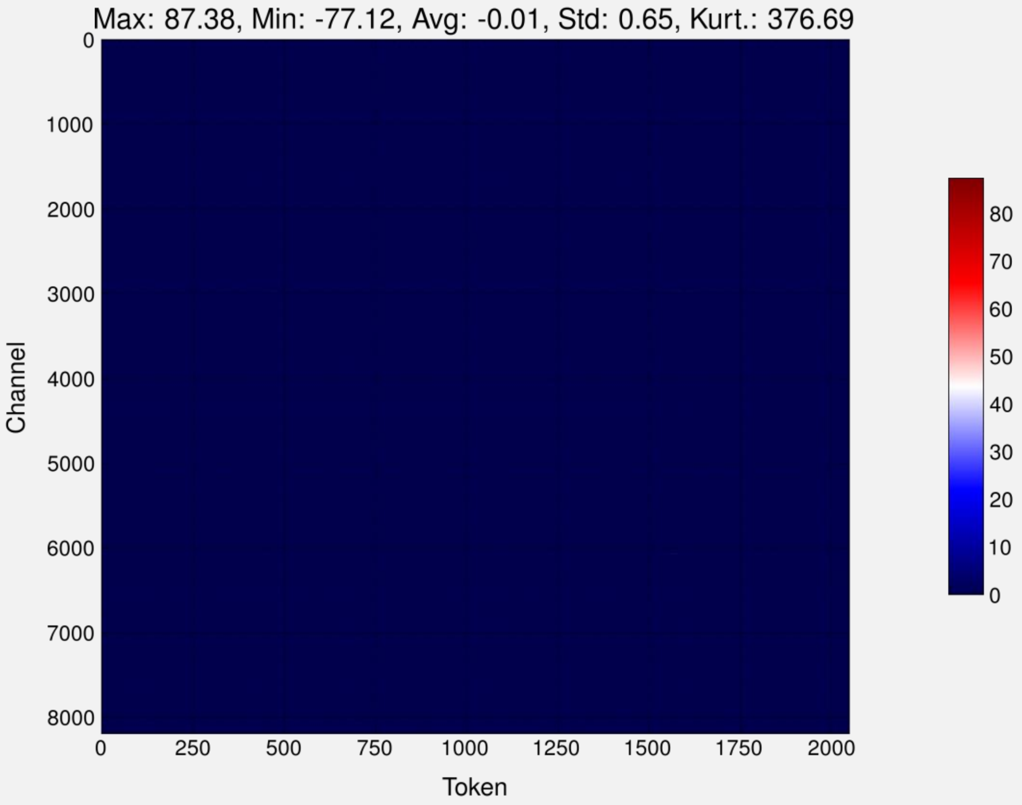
Input distribution w/ and w/o scales

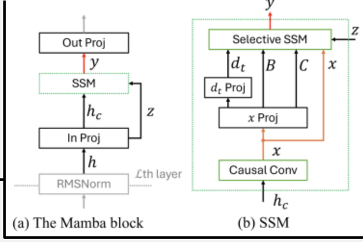
- On a batch, plot of the activations w/ and w/o SINQ scales

Layer: `mamba_layer_63_out_proj_input`



Layer: `mamba_layer_63_out_proj_input_SCALED`





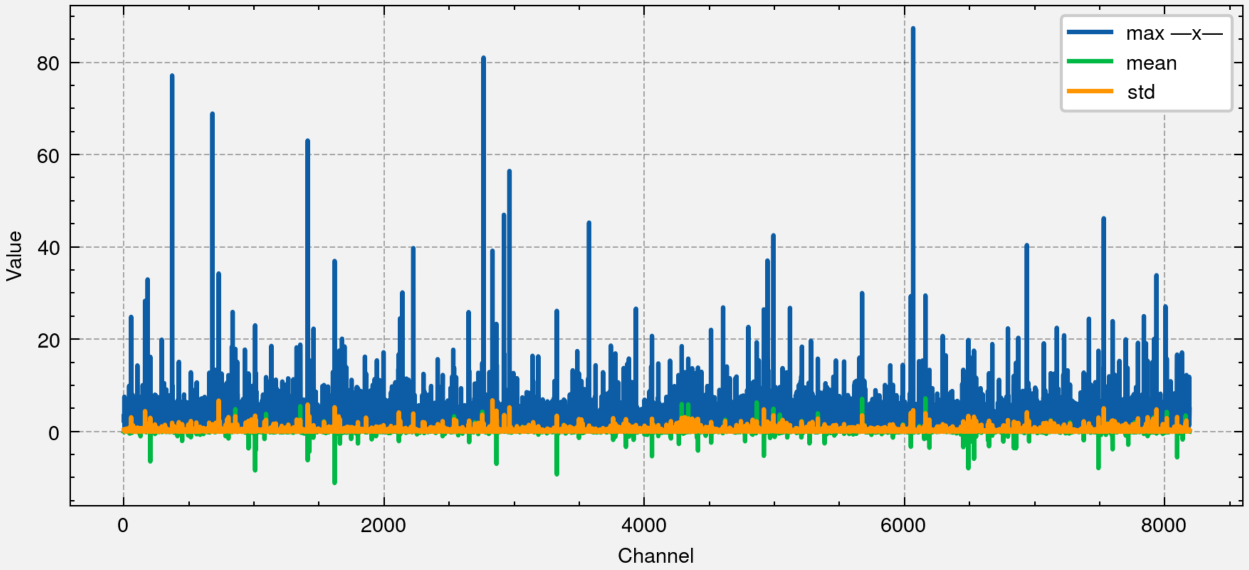
Input distribution w/ and w/o scales

- On a batch, average over the token positions (t)

Layer: `mamba_layer_63_out_proj_input`

Max: 87.38, Min: -77.12, Avg: -0.01, Std: 0.65, Kurt.: 376.69

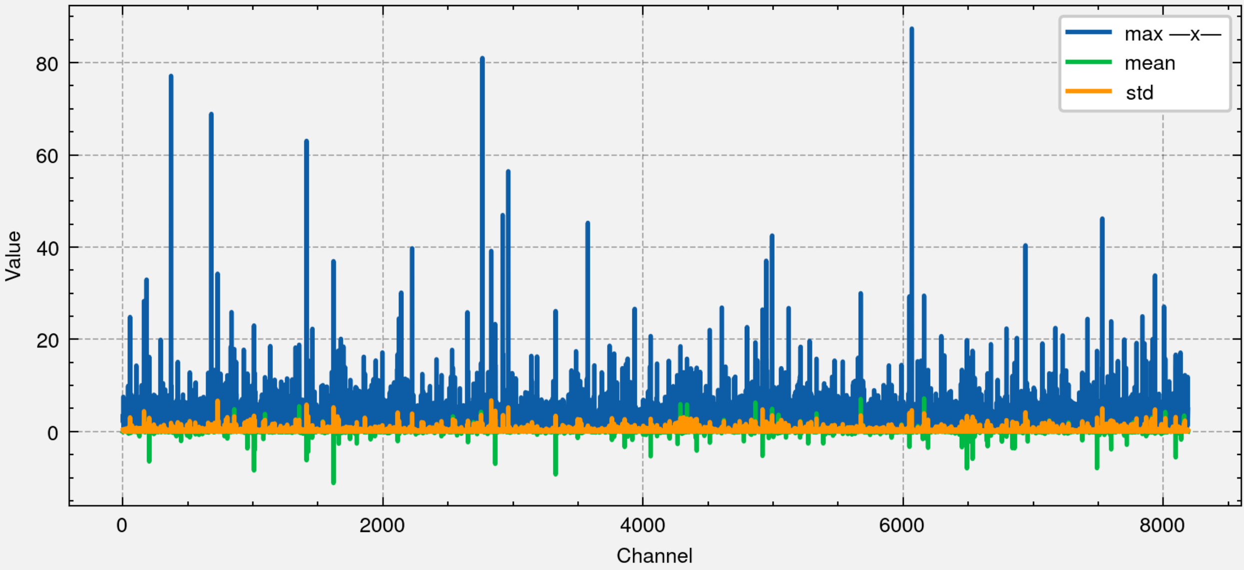
mamba_layer_63_out_proj_input: per-channel activation statistics

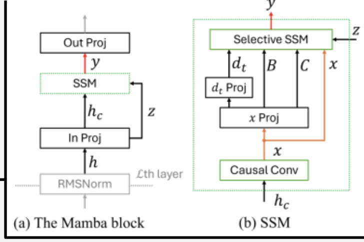


Layer: `mamba_layer_63_out_proj_input_SCALED`

Max: 87.38, Min: -77.12, Avg: -0.01, Std: 0.65, Kurt.: 376.69

mamba_layer_63_out_proj_input_SCALED: per-channel activation statistics





Input distribution w/ and w/o scales

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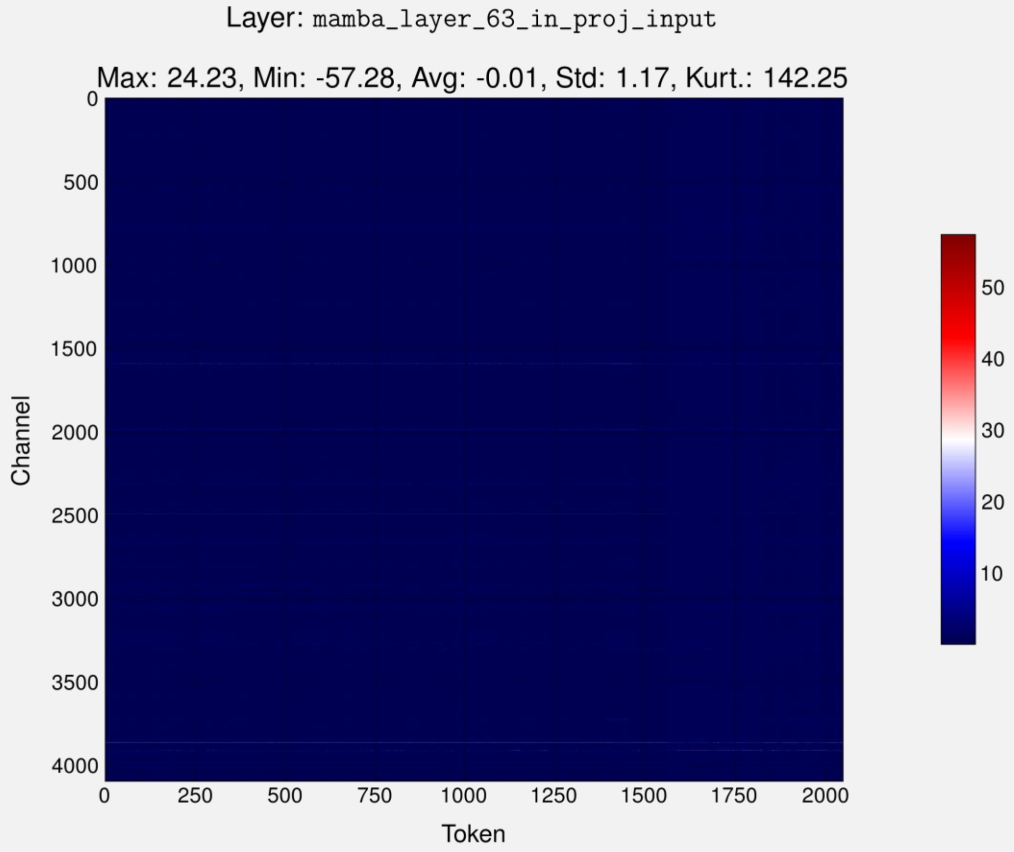

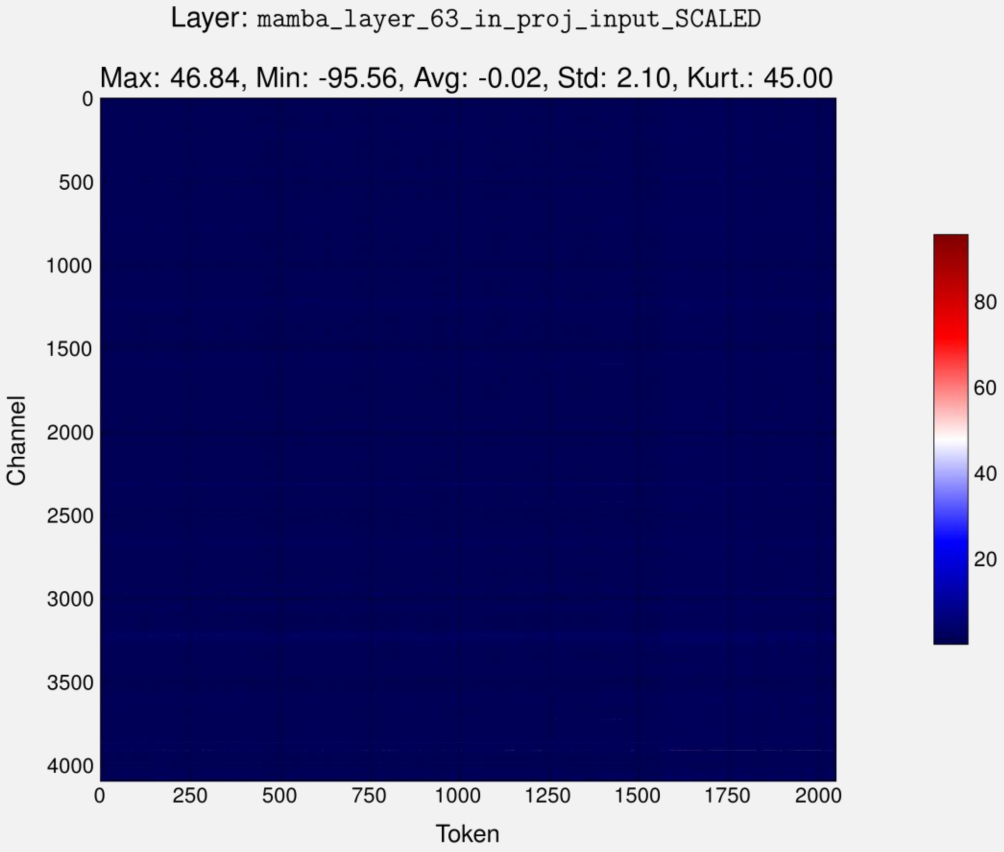

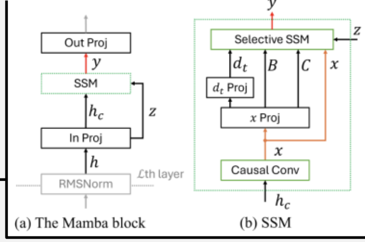

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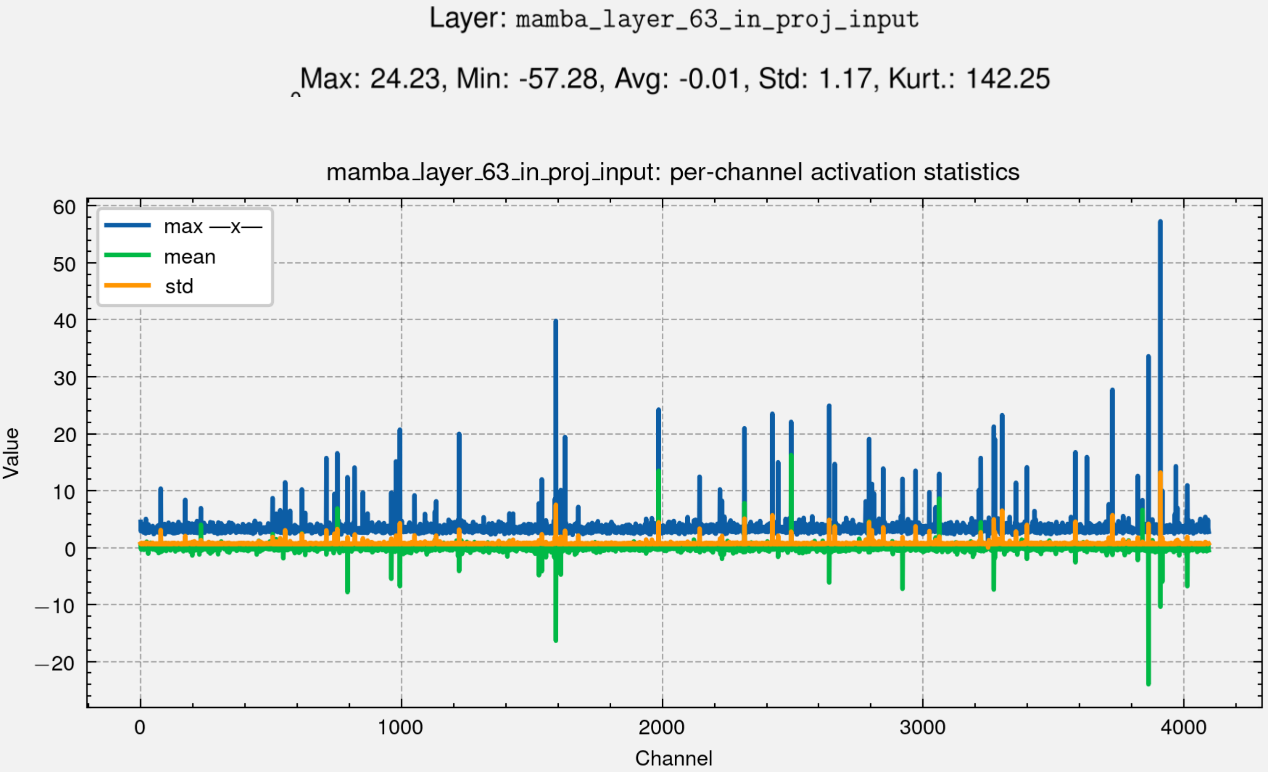

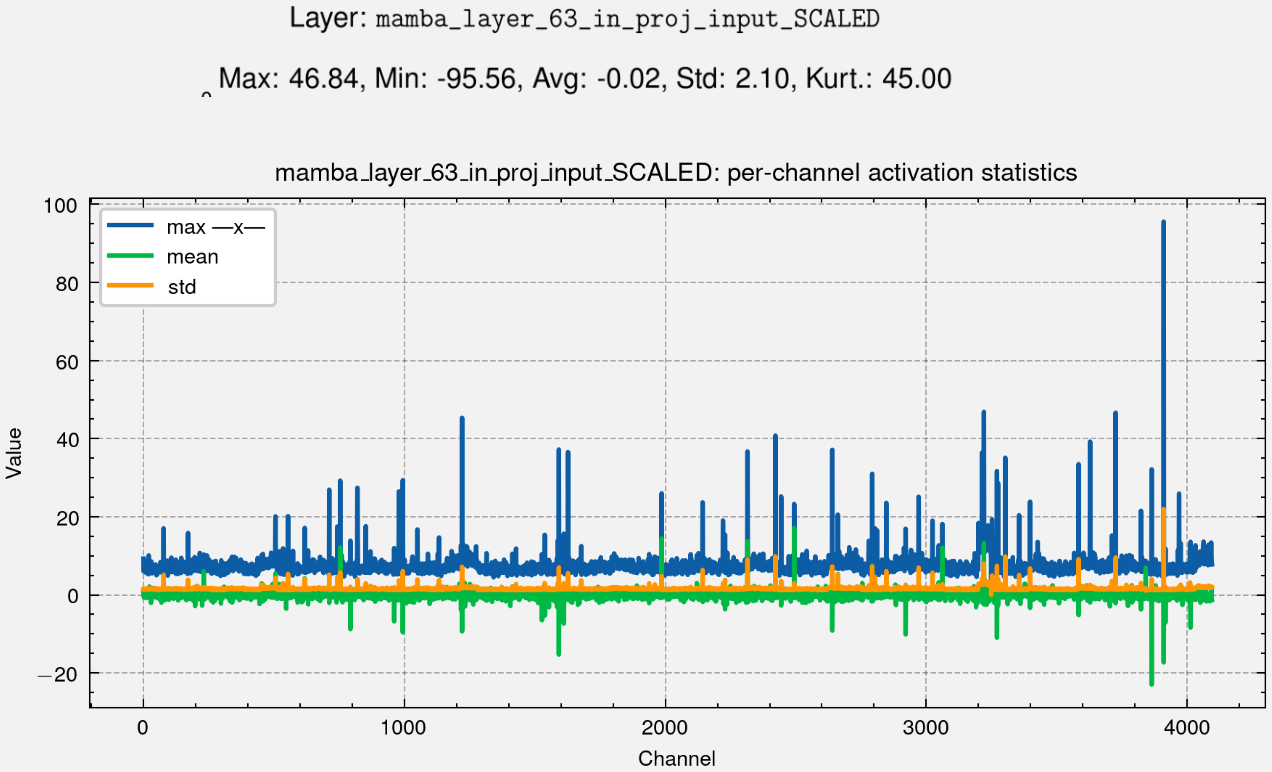

Next steps

- State distribution
- Quamba2